

TAPU CHANDRA BORMON

Dhaka, Bangladesh

☎ (+880) 195-2387780 ✉ tapu.chb99@gmail.com [in LinkedIn](#) [GitHub](#) [Portfolio](#)

Education

Uttara University

2019 – 2023

Bachelor of Science in Computer Science and Engineering

CGPA: 3.70 (out of 4.00)

Research Experience

Research Assistant

2024 – Present

Supervisor: Md. Harun-Ar-Rashid, Senior Lecturer

Uttara University, Bangladesh

- Conducted literature review and assisted in research methodology design under supervisor guidance.
- Participated in experimental setup, data preprocessing, and quantitative result analysis.
- Contributed to manuscript preparation, review, and revision under supervision, supporting overall research quality.

Publications & Manuscripts

IoT-Based Remote Health Monitoring System for Medical Imaging Devices. [Accepted, Springer Nature ITSS-IoE 2025]

Arduino, Raspberry Pi, Python, IoT Protocols (MQTT/HTTP), RTMP, Cloud Storage, DICOM

- Contributed to transforming the initial project into a publishable research paper by restructuring the research framework, refining and articulating the methodology, drafting major sections of the manuscript, and preparing the complete version for successful conference submission as co-author.

Fruit Ripeness Classification Using Computer Vision on Hybrid ResNet50 and SVM Architecture [Conditionally Accepted, IEEE ICEFronT 2026]

Python, TensorFlow, Scikit-learn, ResNet50, SVM, PCA, GridSearchCV

- Developed a hybrid computer vision pipeline integrating ResNet50-based deep feature extraction with SVM classification, enhanced feature representation using PCA, and optimized model performance through k-fold cross-validation; contributed as second author.

Wavelet Packet Decomposition and Cross-Correlation Analysis for Improving ECG Signal Classification. [Under Review]

MATLAB, Python, MIT-BIH Database, Long-term database, ST Change Database

- Developed signal processing pipeline, implemented feature extraction using Wavelet Packet Decomposition (WPD), validated performance on benchmark datasets (MIT-BIH, LTST, European ST Change DB), and contributed as first-author.

Projects

Reddit CTI-NLP Framework | *Python, PyTorch, Transformers, scikit-learn, spaCy, PRAW, Selenium*

[GitHub](#)

- Eliminated data collection downtime by auto-switching to a Selenium scraper whenever the Reddit API failed.
- Cut manual triage effort by fine-tuning DistilBERT to auto-classify CTI-relevant posts from raw noise.
- Reduced false-positive IOCs by combining regex, spaCy, BERT NER, then cross-validating via VirusTotal/OTX.

Stage Forge 3D (SF3D) | *React, Three.js, React Three Fiber, Zustand, WebGL*

[Live Demo](#) | [GitHub](#)

- Implemented demand rendering, cutting GPU load by 80% and eliminating hardware overheating.
- Built a node-based truss connection system with raycast isolation for precise 3D snap-point selection.
- Developed a browser-based CAD-like editor with scene hierarchy, live properties panel, and two-way viewport sync.

E-commerce Website | *React.js, Node.js, Express.js, Redux, MongoDB, PayPal API*

[Live Demo](#) | [GitHub](#)

- Implemented JWT authentication with role-based authorization and protected REST APIs.
- Built a Redux-powered shopping cart and checkout workflow with optimized state management.
- Integrated secure PayPal payments and developed an admin dashboard for products and orders.

Awards & Honors

- Dean's Honor Award, Uttara University, Department of Computer Science and Engineering, Dhaka, Bangladesh (2021)

Technical Skills

Languages: C, C++, Python, R, JavaScript (ES6+), MATLAB, SQL

Web Development: HTML5, CSS3, jQuery, React, Redux, Next.js, EJS, Bootstrap, Tailwind CSS, Vite, WebRTC, Socket.io

Backend & Databases: Node.js, Express.js, MongoDB, Mongoose, MySQL, RESTful APIs, JWT

AI & Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, Matplotlib, NLP (DistilBERT), CNN

Tools & Platforms: Git, GitHub, Postman, Figma, VS Code, Render, npm

Others: Latex, Linux/Unix (terminal & scripting), Cybersecurity

Language Proficiency

IELTS Proficiency Level: B2

IELTS Overall Band Score: 6.5 (Listening: 7.5, Speaking: 6.5, Reading: 6.0, Writing: 6.0)

References

Md. Harun-Ar-Rashid

Senior Lecturer, Department of Computer Science and Engineering, Uttara University

Email: harashid@uttarauniversity.edu.bd

Md. Shafiul Alam Chowdhury

Associate Professor, Department of Computer Science and Engineering, Uttara University

Email: shafiul.cse@uttarauniversity.edu.bd